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Automated Grading System Using NLP For Student Exams

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Abstract
The advancement of Natural Language Processing (NLP) has significantly influenced various educational technologies. This system explores the development of an NLP-based automated grading system that can process handwritten answer sheets uploaded by students. The system utilizes Optical Character Recognition OCR techniques to convert handwritten responses into machine-readable text, followed by NLP algorithms to evaluate the syntactic and semantic accuracy of answers. The system further calculates individual marks, assigns grades based on performance, and ranks students accordingly. This system aims to address the challenges of manual evaluation, such as time consumption, human bias, and scalability, especially in institutions handling large volumes of answer sheets. The automated systems offer efficiency, fairness and scalability. This paper examines existing methodologies including rule-based, machine learning, and deep learning models, with a focus on their application in syntactic and semantic evaluation. Various similarity measures such as cosine similarity, Jaccard index, and semantic embeddings and architectures including BERT, LSTM, and transformer-based models are analysed. This system demonstrates how combining OCR and NLP can revolutionize academic assessment by making it faster, fairer, and more consistent.

Keywords: NLP, OCR, Cosine Similarity, Jaccard Index, BERT, LSTM.

INTRODUCTION

Manual grading of handwritten answer sheets is one of the most resource-intensive processes in educational institutions. Teachers and evaluators often spend significant time evaluating responses, and human scoring is susceptible to inconsistencies, fatigue, and subjectivity. The automated grading system is a solution for this problem. The combination of Optical Character Recognition (OCR) and Natural Language Processing (NLP) offers an innovative way to automate the grading of handwritten descriptive answers. By integrating OCR for handwriting recognition and NLP for textual evaluation, it becomes possible to extract content from studentwritten responses, analyse answer quality based on keyword relevance, grammar, coherence, and semantic similarity with model answers, and assign marks automatically. The system not only grades individual answers but also calculates overall grades and ranks the students within a class based on total scores.

LITERATURE REVIEW

In the paper, Gawande et al. [1] it proposes a conceptual framework for using Natural Language Processing (NLP) to automate the grading of student responses. The authors explore how NLP can improve

objectivity, consistency, and efficiency in academic assessment systems. The study identifies the limitations of manual grading, such as subjectivity, inconsistency, and the time-consuming nature of evaluating large volumes of student responses. To address this, the authors outline a multistage conceptual model that uses NLP techniques to process and grade answers automatically. The model includes key stages that includes text preprocessing, feature extraction and evaluation module. In text preprocessing it includes tokenization, stop word removal and stemming. The feature extraction is done using similarity metrics and semantic analysis. Then it compares the student responses to model answers using the similarity measures. It uses the integration of semantic similarity methods such as vector representations (e.g., Word2Vec or BERT)—to compare the meaning of student responses with standard answers rather than relying purely on keyword matches. The paper emphasizes the importance of contextual understanding in grading, which traditional keyword-based systems often miss. The proposed approach aims to create a system that is scalable, domain-adaptable, and capable of giving consistent feedback.

The author Awaida et al. [2] presents the development of an Automated Arabic Essay Grading System that utilizes Arabic WordNet and evaluates performance using the F-score as a key metric. The aim of the study is to enhance the accuracy and fairness of grading Arabic-language essays by employing semantic analysis rather than relying solely on keyword or surface-level features. This proposed system also includes the preprocessing stage, semantic analysis and similarity scoring. The essays are cleaned and normalized, with stop-word removal, tokenization, and stemming tailored for Arabic text processing. The Arabic WordNet is employed to detect semantic similarity between student-written essays and model answers. This allows the system to consider different valid expressions and synonyms that may be used by students. The core scoring is based on the precision, recall, and Fscore are the common metrics in information retrieval and classification tasks. The dataset used in this are of Arabic student essays and model answers is used for evaluation.

Moizuddin et al. [3] introduces a machine learning-based system for automatically scoring English essays. The goal of the system is to reduce the manual workload in educational assessment and to provide faster, unbiased, and consistent grading for written English content. The proposed system follows a structured approach comprising the following key components such as data collection and preprocessing, feature extraction, model training, evaluation and system output. The text data is cleaned, tokenized and vectorized using NLP techniques. The model is trained using machine learning algorithms that includes linear regression, SVM, Random Forest and logistic regression. The models are evaluated using performance metrics like Mean Squared Error (MSE) and R-squared. The paper finds that the Random Forest algorithm performs best in predicting scores closely aligned with human-assigned grades. The paper uses an existing dataset of English essays scored by human raters. The features include the syntactic, semantic and statistical elements such as TF-IDF vectors, sentence length, word count and grammatical correctness. The study concludes that machine learning techniques, especially Random Forest, can offer robust performance in automatic essay scoring for the English language.

Abdulsamad et al. [4] build a system for automatic grading of Arabic short answer questions, aimed at improving the efficiency and objectivity of evaluating student responses in Arabic. The motivation behind the study lies in the manual grading challenges unique to Arabic, including morphological richness, word variation, and syntactic complexity. The authors designed a system that compares student answers to model answers using text similarity techniques and NLP tools tailored to Arabic. The process of this involves several stages that includes preprocessing where the Arabic text is normalized. The system calculates similarity between student and reference answers using cosine similarity, Jaccard similarity, TF-

IDF and word matching techniques. The system was tested on a custom-built dataset of Arabic short answers.

Ghelani et al. [5] presents a comprehensive review of various automatic student grading systems, highlighting the evolution, methodologies, technologies, and challenges involved in automating the evaluation process across educational domains. The paper focuses on analysing existing approaches for automatic grading, primarily for objective, short answer, and essay-type questions. It explores both traditional rulebased systems and modern AI-driven models. In this review paper the study does not use a specific dataset.

Ghadekar et al. [6] published in the Journal of Integrated Science and Technology (2025), proposes a robust and scalable solution to a long-standing challenge in education. It reduces the burden placed on educators by time-consuming. It is a comprehensive system for automatically grading handwritten examination scripts using a combination of Optical Character Recognition (OCR) and Machine Learning (ML) techniques.

The process begins by converting scanned PDFs of handwritten answer sheets into machine-readable arrays. Preprocessing techniques such as grayscale conversion, binarization, noise removal, and segmentation are applied to enhance the accuracy of OCR in recognizing diverse handwriting styles. After digitization, the system performs structured question-answer mapping, aligning each student's response with the corresponding question section. This is done to preserve contextual understanding and enable section-wise evaluation. If model answers are available, the system uses them as benchmarks for comparison. In such cases, responses are graded based on semantic similarity, syntactic structure, grammar correctness, and keyword relevance. It employs Retrieval-Augmented Generation (RAG) to generate appropriate reference answers using relevant databases and information retrieval techniques, making it adaptive and dynamic. It uses evaluation features such as TF-IDF scores, context matching, and grammar checks. The system's evaluation was based on a custom dataset of handwritten student responses, which included both textual answers and diagrammatic content.

Ouahrani et al. [7] it makes a valuable contribution to the field of Natural Language Processing (NLP) and educational technology by introducing AR-ASAG, the first dedicated dataset for Arabic automatic short answer grading (ASAG). The dataset's development fills a significant gap in Arabiclanguage educational resources. It also proposes an unsupervised grading method using semantic similarity techniques, particularly COALS-based word embeddings, combined with TF-log weighting and syntactic overlap metrics. It presents a well-executed and impactful effort to bridge the gap in Arabiclanguage educational NLP tools. By introducing a high-quality dataset and demonstrating a functional unsupervised grading approach, the authors contribute significantly to both research and practical implementation in Arabic ASAG.

Mandal et al. [8] this paper presents a novel method for recognizing handwritten English characters using a technique called Column-wise Segmentation of Image Matrix (CSIM), implemented within an Artificial Neural Network (ANN) framework. This improves the performance of character recognition systems by reducing data dimensionality and capturing distinct features across handwriting styles. It begins with scanning characters written within 80×80-pixel boxes. These images are binarized and compressed into 10×10 matrices by dividing them into uniform 8×8 blocks and assigning a binary value based on the presence of ink (1 or 0). The compressed matrix is then segmented column-wise into 10 columns, and each column is treated as a feature vector input to the ANN. It is trained using the Perceptron Learning Algorithm. The column-wise segmentation improves character differentiation compared to previous row-

wise methods. It provides a more efficient and accurate method for recognizing handwritten English characters. The authors conclude that CSIM is faster, simpler, and more accurate than previous methods.
In this paper, Das et al. [9] it presents a well-structured and technically sound approach for improving the recognition accuracy of offline handwritten English characters using Hidden Markov Models (HMM). The authors integrate both global and local feature extraction techniques, combining gradient, curvature, and projection features with a segmented blockbased analysis for localized details. The method is further strengthened by the use of multiple HMMs for characters with high structural variation, and a post-processing stage to resolve confusion between similar-looking characters. Here in this paper, the authors address the limitations in recognition accuracy caused by the variability in handwritten character shapes, sizes, and writing formats. To overcome this, they introduce a hybrid feature extraction technique. The HMM framework is trained using the Baum-Welch algorithm, and recognition is performed using the Viterbi decoding algorithm. The dataset used in this comprises 13,000 samples collected from 100 writers, each contributing five instances per character. The results show that the proposed system achieves an average recognition accuracy of 98.26, with significant improvements when using the multi-HMM and post-processing enhancements.
Badry et al. [10] introduces an Automatic Arabic Grading System for Short Answer Questions (AASAG) designed to evaluate student responses in Arabic with high accuracy and minimal human intervention. The system compares student answers to instructor-provided model answers using semantic vector space representations. The grading process is divided into three core stages: text preprocessing, term weighting, and semantic similarity computation. Preprocessing includes tokenization, normalization, lemmatization, and stop-word removal, tailored specifically for Arabic using tools such as Farasa and MADAMIRA. The system is evaluated using the AR-ASAG dataset, which comprises 2133 student-model answer pairs on Arabic exam questions. The paper suggests future improvements using deep learning, its current contribution is valuable for Arabic-language educational systems.
Ayaan et al. [11] proposes a sophisticated automated grading system designed to evaluate open-ended short-answer responses using a hybrid of Natural Language Processing (NLP) techniques and semantic analysis. The core aim is to improve grading consistency and reduce educator workload by automating the evaluation process, which is traditionally timeconsuming and subjective. The system works by comparing student responses to reference answers using multiple layers of similarity measures. These include edit similarity, cosine similarity, Jaccard similarity, and normalized word count, each weighted differently. The outputs of these measures are combined into a base NLP score. TensorFlow's Universal Sentence Encoder is used to compute semantic similarity, which captures deeper contextual understanding of student answers beyond surface-level text. The final grade is computed through a rule-based system that considers both the semantic and NLP scores. If semantic similarity is low, the answer is automatically scored zero. High semantic similarity and word count yield full marks. Visualization tools such as radar charts were used to display scoring outcomes.
Pramit et al. [12] presents an Automated Essay Grading System (AES) that uses Natural Language Processing (NLP) techniques to assess student essays efficiently and fairly. The motivation behind the study stems from the limitations of existing AES models, particularly their inability to effectively evaluate semantic aspects of essays and adapt to different rubric structures. The proposed system grades essays using two categories of features : Syntactic features and Semantic features. The grading methodology involves preprocessing essays to clean and normalize the content, followed by a multi-step scoring approach using custom rule-based logic and machine learning models. , the authors developed a

bidirectional LSTM model trained on essay datasets to assess writing style and semantic content. The study says that while the system performs well and is adaptable for schoollevel essays, it requires further refinement and more diverse datasets to be fully reliable in broader academic contexts. Future work includes improving dataset quality, enhancing the semantic analysis, and developing domain-independent models for robust grading.

Gabon et al. [13] paper presents the development and evaluation of an automated essay grading (AEG) system using Natural Language Processing (NLP) techniques. The authors aim to determine how closely the automated system can replicate human grading across five types of essays that are persuasive, descriptive, narrative, argumentative, comparison and contrast. The system is designed in two phases essay evaluation and model training. It uses the preprocessing techniques like lemmatization, tokenization and stop-word removal. The data for this system was collected from 35 IT students. The results are compared to the system's output using Weighted Cohen's Kappa and the Friedman test. The paper concludes that the AEG system is a scalable and efficient tool for educational assessment. It supports instructors by reducing grading workload while maintaining fairness and consistency. Future work involves expanding the dataset, incorporating deeper semantic features like creativity and tone, and ensuring demographic fairness across varied student populations. Sadanand et al. [14] presents a robust approach for automating the grading of student essays using advanced machine learning and NLP techniques. The system is developed to reduce the inefficiencies and subjectivity in manual essay grading while offering real-time, unbiased, and accurate feedback to students. It uses a Long Short-Term Memory (LSTM) neural network model trained on a dataset of 12,000 manually graded essays sourced from the Hewlett Foundation. The grading model is based on the factors on factors such as quality of content, grammar and spelling (syntactic) errors, sentiment, and plagiarism. The system uses sentiment analysis using a Na¨ıve Bayes classifier trained on a Twitter dataset to determine the tone of the essay (positive, negative, or neutral). The system achieved a high performance with a Quadratic Weighted Kappa (QWK) score of 0.911, indicating strong agreement with human raters, and 99.4 accuracy in sentiment analysis. The system used in here improves the students writing skills.

Ibrahim et al. [15] proposes the development of an Artificial Intelligence based automated essay grading system to address the challenges associated with manual essay assessments in educational institutions, particularly in Nigeria. The system architecture incorporates modules for document vector creation that contains the question and student response, comparison material, feature extraction, scoring model creation, and score assignment. This system is implemented using Java and MySQL. The system evaluates the performance using Pearson correlation. The system is limited for handling short text answers and the paper concludes by recommending further research to expand the grading capabilities to support a wide range of question types.

Wilson et al. [16] presents an automated essay scoring (AES) system that leverages multi-model machine learning techniques combining traditional feature extraction and deep learning with word vector models to evaluate essays with nearhuman accuracy. The system uses feature extraction techniques to gather word count, grammar mistakes, sentence structures, and part-of-speech (POS) counts using the spaCy library. The semantic features are captured using GloVe word embeddings to create contextual representations of essay content. Three types of neural networks such as LSTM,GRU and BiLSTM were tested with varying GloVe embedding sizes. The limitations of using this system includes it gives lower performance on longer essays. The authors proposes TF-IDF in future iterations for improving performance. Rahman et al. [17]: proposes a system that automatically extracts text from scanned answer scripts using Optical Character

Recognition (OCR) . It summarizes the content, preprocesses it, and calculates several similarity metrics against model answers to generate scores. The system focuses on five core parameters for scoring: synonym similarity, bigram similarity, cosine similarity, Jaccard similarity, and grammatical/spelling errors. The text summarization is conducted using a keyword-based technique, which outperformed the bag-of-words approach in F-score metrics. After preprocessing with NLTK, they handles the tasks like tokenization, stopword removal, and lemmatization are calculated. The paper demonstrates how NLP techniques can enhance the accuracy, efficiency, and fairness of academic assessments, marking a significant step towards intelligent educational technologies.

Mahana et al. [18] explores the design and implementation of an automated essay grading system using machine learning techniques, focusing on predicting human-like scores for essays. The dataset is used from Kaggle. The system architecture is built around linear regression trained on extracted features such as bag of words, POS ,spelling error counts and punctuation. A 5-fold cross-validation, is used for evaluating the model across all essay sets using the Quadratic Weighted Kappa (QWK) metric to measure agreement between machine and human scores. The authors show that linear regression models, when used with meaningful NLP-based features, can replicate human-like grading with reasonable reliability.

Shehab et al. [19] focuses on developing an efficient and reliable system for automatic grading of Arabic essay-type answers. This research investigates both string-based and corpus-based similarity algorithms to automate the grading process for Arabic essays. Four algorithms are analyzed and compared that are Damerau-Levenshtein which is a stringbased method measuring the minimal edit distance between two strings. N-Gram another string-based technique analyzing the common subsequences shared between model and student answers. Latent Semantic Analysis a corpus-based method that builds a semantic space to represent word meanings. DISCO2 an another corpus-based method relying on distributional similarity derived from co-occurring word contexts. Preprocessing steps such as tokenization, stopword removal, and stemming are employed to refine the input data. In conclusion , the authors recommend future work combining both string- and corpus-based algorithms to further improve grading accuracy.

This paper by Suresh et al. [20] explores the application of advanced AI and natural language processing models in automating educational assessments. The system processes textual input to generate context-aware questions, evaluates student responses using semantic similarity measures, and adjusts question difficulty in real-time based on student performance. It reviews the limitations of traditional rule-based and machine learning approaches. The system faces challenges such as high computational cost, potential bias from pretrained models, and the need for high-quality input data. The paper contributes significantly to the domain of AI in education, showcasing how modern NLP models can revolutionize assessment systems by making them more intelligent, adaptive, and learner-centric.

CONCLUSION

Automated grading systems using NLP have shown significant promise in transforming traditional educational assessment. By combining robust handwriting recognition with intelligent text analysis, the system has the potential to drastically reduce the burden on educators, minimize grading bias, and ensure faster feedback to students. Furthermore, the integration of grade assignment and class ranking functionality offers an end-to-end assessment solution suitable for schools, colleges, and universities. The review highlights that while current models offer substantial improvements in efficiency and consistency, they still face limitations in understanding nuanced language, domain-specific context, and diverse student expression.

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